DATA SCIENCE AND VISUALIZATION

WEEK 1 – TEAM FORMATION & DATA HUNTING

Project: Youth Smoking and Drug Consumption: From Raw Data to Visual Insights

1. Introduction

This document records the team formation and dataset selection activities for the Data Science and Visualization project. The project explores demographic, personality-related, and substance-consumption patterns in the selected dataset.

2. Team Details

Team Number: 16

Course: Data Science and Visualization

Course Code: 24ADI204

S. No.	Team Member	Register Number
1	Lakshmi Prabha M M	25BAD049
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3. Project Title

Youth Smoking and Drug Consumption: From Raw Data to Visual Insights

4. Problem Statement

Smoking and substance consumption are important public-health topics. The selected dataset contains demographic information, personality measurements, and responses related to the consumption of different substances. Understanding patterns across these variables requires systematic data exploration and clear visualizations. This project aims to examine the dataset, identify observable patterns and associations, and communicate findings through visual analysis. The analysis is exploratory and does not establish cause-and-effect relationships.

5. Project Objectives

- Understand the structure and characteristics of the selected dataset.
- Identify and handle data-quality issues during data preparation.
- Explore demographic and substance-consumption patterns using summaries and visualizations.
- Examine associations between personality measurements and substance-consumption responses.
- Apply suitable feature-engineering and dimensionality-reduction techniques.
- Present findings through an interactive dashboard.

6. Dataset Selection

Dataset name: Drug Consumption UCI (Missing Values)

Source platform: Kaggle

Dataset link: https://www.kaggle.com/datasets/thelonestar033/drug-consumption-uci-missing-values

The dataset includes demographic attributes, personality-related measurements, and consumption-response variables for multiple substances. These fields support exploratory analysis and visualization.

7. Reason for Selecting the Dataset

- It contains demographic, personality, and substance-related variables.
- It supports univariate and bivariate exploratory data analysis.
- It provides opportunities to practise data cleaning, feature engineering, and dimensionality reduction.
- It allows the team to explore patterns across different groups and measurements.

8. Planned Project Workflow

1. Understand the dataset and its variables.
2. Inspect and clean missing or inconsistent data.
3. Perform exploratory data analysis and create visualizations.
4. Engineer and transform selected features.
5. Apply dimensionality-reduction methods where appropriate.
6. Build a dashboard to communicate the results.

9. Week 1 Outcome

The project team, project topic, and dataset source have been identified for the subsequent stages. The dataset will be examined in detail during Week 2 before cleaning and analysis activities are documented.

10. Reference

Kaggle – Drug Consumption UCI (Missing Values):

https://www.kaggle.com/datasets/thelonestar033/drug-consumption-uci-missing-values